PROJECT

01. Refer topic 02 in page 02.

02. Specifications

Boost Converter Specifications

Rated Power: ~~1.5kW~~ 1kW
Input Voltage Range: 200 – 400 Vdc
Nominal input voltage: 200 Vdc
Output Voltage: 400 Vdc
Rated output current: ~~3.75A~~ 2.5A @ 200V
Maximum input current at 200VDC: 5A @ 200V
Switching frequency: 100 kHz
Operating Mode: CCM
Inductor ripple: 10 – 20%

$\frac{500}{100}$ $\frac{1000}{200}$

03. Duty cycle

▲ CCM $V_o = \frac{V_{in}}{1-D}$

given
$V_o = 400V$
$V_{in} = 200V$

So ⇒ $400 = \frac{200}{1-D}$

$D = 0.5 = 50\%$ duty cycle

04. Switching

▲ $f_s = 100kHz$

$f_s = \frac{1}{T_s}$ ⇒ $T_s = \frac{1}{f_s}$

$T_s = 0.00001 s$
$= 10 \times 10^{-6} s$
$T_s = 10\mu s$

▲ $D = \frac{T_{on}}{T_s}$

$T_{on} = 5\mu s$
$0.5 = \frac{T_{on}}{10\mu s}$
$T_{off} = 5\mu s$

05. calculation of output /load current

given $P_o = 1kW = 1000 W$
$V_o = 400V$
for DC Voltage
$P_o = V_o I_o$
⇒ $I_o = \frac{1000}{400}$
$I_o = 2.5A$

⇒ $R_L = \frac{V_o}{I_o}$

$R_L = \frac{400}{2.5} = 160\Omega$

06 Inductor Design [Area product method]

01. $f_{sw} = 100kHz$
$L = 1000\mu H < L < 2000\mu H$
$I_{L,RMS} =$
$\Delta i_L =$
$I_{L,avg} \approx \frac{P}{V_{in}} = \frac{1000}{200} = 5A$
$L\,ripple = 10 - 20\%$

▲ ⇒ "L" from $\Delta I_L = \frac{V_{in} D}{L f_s}$ ⇒ $L = \frac{V_{in} D}{\Delta I_L f_s}$
CCM

⇒ 10% ripple case (ΔI_L)

$\Delta I_L = 0.1\, I_{L,avg}$
$= 0.1[5]$
$\Delta I_L = 0.5A$

⇒ Peak Ind, current $I_{L,max}$

$I_{L,max} = I_{L,avg} + \frac{\Delta I_L}{2}$
$= 5 + \frac{0.5}{2}$
$I_{L,max} = 5.25A$
$I_{L,min} = 4.75A$

⇒ $L = \frac{V_{in} D}{\Delta I_L [f_s]}$
$= \frac{200[0.5]}{0.5[100 \times 10^3]}$
$= 2mH$
$= 200 \times 10^{-5} H$
$= 2000\mu H$

⇒ 20% ripple case (ΔI_L)

$\Delta I_L = 0.2 I_{L,avg}$
$= 0.2[5]$
$\Delta I_L = 1A$

⇒ (Peak Inductor) $I_{L,peak}$

$I_{L,peak} = I_{L,avg} + \frac{\Delta I_L}{2}$
$= 5 + \frac{1}{2}$
$I_{L,peak} = 5.5A$ or $I_{L,max}$

⇒ $I_{L,min} = I_{L,avg} - \frac{\Delta I_L}{2}$
$= 5 - \frac{1}{2}$
$I_{L,min} = 4.5A$

⇒ $L = \frac{V_{in} D}{\Delta I_L f_s}$

$L = \frac{200[0.5]}{1[100 \times 10^3]}$
$= 1000\mu H$

$L = 1500\mu H$

⇒ $\Delta I_L = \frac{V_{in} D}{L f_s} = \frac{200(0.5)}{1500\mu \times 100k} = \frac{66666 \times 10^{-4} \times 10^{-3}}{0.6666A}$

⇒ % ripple

$\Delta I_L = x\, I_{L,avg}$
$0.6666 = x\, 5$
$x = 0.1333$
$x\% = 13.33\%$

⇒ Stored energy

$\frac{1}{2} L I_{rms}^2 = \frac{1}{2}[1500\mu][5.333]^2 = 21.33306 \times 10^{-6}$
$= 21.33306\, mJ$

⇒ $I_{peak} = I_{avg} + \frac{\Delta I_L}{2}$

$= 5 + \frac{0.6666}{2}$
$= 5.3333\, A$

$I_{min} = I_{avg} - \frac{\Delta I_L}{2}$
$= 4.6666A$

AREA PRODUCT

Tasks
1. Compute the peak current $I_{L,max}$ and RMS current $I_{L,RMS}$ for the given ripple.

2. Calculate the required area product A_p. Search a manufacturer datasheet/catalog and select a suitable core. Record its relevant parameters (A_c, A_w, path length, volume, mean turn length).
3. Determine the number of turns N and select a standard wire size.
4. Calculate the DC resistance of the winding and the copper loss.
5. Size the required air gap l_g using two independent methods, and cross-check your results.
6. Verify the actual inductance achieved with the rounded number of turns.
7. Estimate total losses (copper + core) and the resulting temperature rise. Check whether the winding actually fits in the core window.
8. If any constraint is violated, revise the design and iterate.

$I_{L,avg} = 5A$ $\quad \Delta I_L = 0.6666$
$I_{L,peak} = 5.3333A$
$I_{L,min} = 4.6666A$

$I_{L,rms} = \sqrt{I_{L,avg}^2 + \frac{\Delta I_L^2}{12}}$

$= \sqrt{[5]^2 + \frac{(0.6666)^2}{12}}$
$= 5.0037$

⇒ $A_p = A_w A_c$ (Ae, Amin)

$A_p = \frac{L I_{L,rms} I_{L,max}}{K_w B_{max} J}$

$= \frac{1.5 \times 10^{-3}][5.0037][5.3333]}{[0.4][100 \times 10^{-3}][2.5]}$

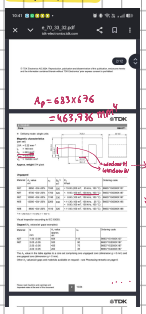
$= 0.312729 \times 10^6\, mm^4$
$= 312,729\, mm^4$

(⇒ A_p check sheet & according to A_p choose on core
⇒ core sheet check B_{max}
⇒ matches fix core)

Units
$H\,A_{rms}\,A$ / $[mWb]\,[T]\,[m^2]$; $\frac{Wb}{A} = H$
⇒ $\frac{H \cdot A}{[Wb]\,[m^2]}$; $Wb = Henry \times A_m$
⇒ $[m^2][m^2]$
⇒ $[10^3 mm^2][m^2]$
$= 10^3 mm^2$; $1m^2 = 10^6 mm^2$
$10^{-6} m^2 = 1mm^2$

$E_{10} = 151,007$

sheet 01 E70



window H 21.4 x 12 = 681.3 mm^2
$A_p\ A_w$
$= 633 \times 481.3$
$= 329,059\ mm^4$

A_w & A_{min}
$= 633 \times 480$
$= 303,350\ mm^4$

TURNS

$N = \frac{L \times I_{L,max}}{B_{max} \times A_c}$

$= \frac{1.5 \times 10^{-3} \times 5.3333}{100 \times 10^{-3} \times 683} 10^6$

$= 1.1712 \times 10^{-4} \times 10^6$
$= 117.12\ turn$

$\frac{H \times A}{\frac{Wb}{m^2}} \times mm^2$
$= \frac{m^2}{mm^2}$
$= \frac{10^6 mm^2}{mm^2}$

∴ based on core select boin
⇒ if it is Toroid no need
∴ $J = 3 - 3.5$
∴ $B_{max} = 100mT$
∴ $K_w = 0.4$